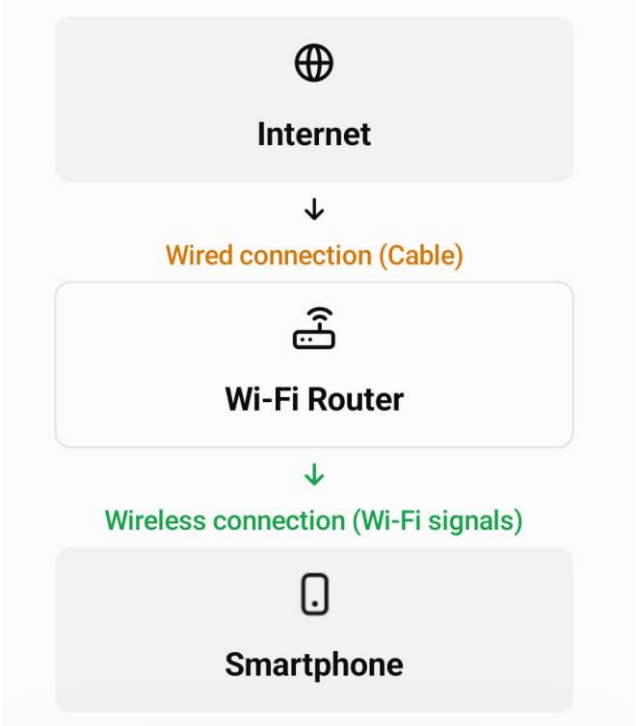
Assignment – 1.11

Wireless Networking

Q - 1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Wi-Fi Standard	Wi-Fi Generation	Frequency Band	Maximum Speed
802.11n	Wi-Fi 4	2.4 GHz and 5 GHz	600 Mbps
802.11ac	Wi-Fi 5	5 GHz	6.9 Gbps
802.11ax	Wi-Fi 6	2.4 GHz and 5 GHz	9.6 Gbps

<u>Q - 2.</u> Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



➢ **<u>Explanation:</u>**

- **<u>Wired connection:</u>** A cable connects the internet service modem to the Wi-Fi router.
- **<u>Wireless connection:</u>** The smartphone connects to the router using Wi-Fi signals.
- The router allows the smartphone to access the internet.

Q - 3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why. Hint: Consider both signal strength and security when choosing.

➢ Wi-Fi Network Scanning:

No.	SSID (Wi-Fi Name)	Signal Strength	Security
1	Home_WiFi	Strong	WPA2
2	Office_WiFi	Medium	WPA3
3	Public_WiFi	Weak	Open

➢ My Choice:

- I would choose Office_WiFi because it has good security (WPA3) and a good signal.

Q - 4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

- **SSID Hiding:**

  SSID hiding means hiding the name of a Wi-Fi network so it does not appear in the normal list of available Wi-Fi networks.

- **Advantage:**

  It makes the Wi-Fi network less visible to other people.

- **Disadvantage:**

  It does not provide strong security because the network name can still be discovered by using special tools.